Hydro geochemical Studies of Nambulapulakunta Mandal, Anantapur District, Andhra Pradesh by using Geospatial techniques

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Abstract

Hydro geochemical studies were conducted by using geospatial techniques in the current study area of Nambulapulakunta Mandal, Anantapur district, AP, India. Usage of saline water for irrigation is the best alternative for overcoming the constraints and increasing crop yields. For analysis, ground water samples are collected from 22 locations of the study area. The assessment of quality of groundwater for the purpose of drinking and irrigation was carried out using pre-post monsoon water samples. The major parameters analyzed for this purpose are pH, EC, TDS, carbonate, Bicarbonate, alkalinity, and chloride. Presence of elements like potassium, magnesium, sodium and calcium were also analysed. Sodium adsorption ratio of the sample were also analyzed. Based on Gibbs' ratio, samples collected in both of the seasons belongs to rock dominance field. Piper diagrams prepared for hydro geochemical classification and hydro chemical processes supports that most of the collected water samples belongs to the category of (Na + K) / (Na + K + Ca) and Cl / (HCO_3 + Cl). Analyzed Geochemical data were evaluated and compared with WHO water quality standards. All parameters are within limits of WHO hence, useful for dinking purposes. Water quality analysis of various parameters conducted in the study area show that the ground water of the area is suitable for both drinking and irrigation.

Keywords: Groundwater, Hydro geochemistry, Geospatial techniques, Irrigation, TDS.

Introduction

The fortune of subsurface contamination depends on the local geology, groundwater drainage patterns, pore-scale processes, and molecular-scale processes. Contamination strength reaches quickly within a high-conductivity sand lens, or it might diffuse at a snail's pace through low-conductivity clay. A number of contaminants absorb onto the surface of aquifer solids, moving very little from their source, at the same time as others travel liberally through the flowing pore water, sometimes ending up lots of kilometers from their source. Chemical reactions alongside the way know how to reason a contaminant to disappear, or worse, appear from apparently nowhere. When we speak of groundwater contamination that means solutes are dissolved in the water that can cause to be it unsuitable for drinking purposes. Mostly natural waters hold at the smallest amount of dissolved substances it is reflecting of as contaminants. For example, water glasses are consisting a little of lead and arsenic, Resulted, no risk for humans because these substances are very low concentrations presented. But, where are increases there is some significant risk to human health or an ecosystem (Aghazadeh et al., 2010).

Ground water is the major resource of drinking water. The ground water is clean and free from pollution when compared to surface water. The ground water is polluted due excessive use of fertilizers and pesticides, increased human activities and rapid growth of industries (Pradeep Kumar et al., 2020). Since Ground water contains high amount of various Ions, salts etc. its quality has to be tested before using it as portable water for domestic purposes. Otherwise it may lead to water borne diseases. Hence it is very essential to maintain the quality of ground water for human consumption. Present study was under taken to investigate the impact of the ground water quality of Bore wells water samples in Anantapur at various locations. Thus in this work the physico-chemical parameters of ground water like pH, Temperature, EC, TDS, DO, Alkalinity, Hardness, Calcium, Magnesium, chlorides, etc. Continuous discharge of industrial waste, domestic sewage and solid waste cause the ground water to be polluted. Since the ground water is continuously polluted it is necessary to protect and manage the ground water quality.

Analysis of remotely sensed data for drainage, geological, geomorphological characteristics of terrain in an integrated way facilitates effective evaluation of groundwater potential zones. Similar attempts have been made in the generation of different thematic maps for the delineation of ground potential zones in a different parts of the study area. (Doneen et al., 1964; Freeze et al., 1979; Gibbs et al., 1970; Hem et al., 1989;). A total of three thematic maps such as geological, geomorphological, and hydrological maps were prepared based on image interpretation studies with limited field checks and analysis of available data. The geomorphology map depicts the various landforms evaluate through a timely geomorphic process and is a basic input to evaluate resource potential associated with the landforms. The hydrological map provides a basis for potential and non - potential areas for groundwater development based on geomorphological, geological, and structural information.

Study Area

The study area is located on 78° 45'0" to 78°15'32"E longitude 13°55'0" to 14° 10' 0" N latitude with intended boundary falling in survey of India (SOI) topographic sheet 57 J/08and 57K/05 on 1:50,000 scale (Figure 1) of the area covered is about 120 km. Eastern and S-W part of the study area is occupied by high hills, ridges, and valleys. The Papaghni river in this study area is a seasonal river fed by monsoon (June – October) which is following in the S-N direction. The tanks, water bodies, lakes are covered randomly in the study area. The climate is dry with a mean annual rainfall of 50-60 cm and a mean annual temperature of 32^0c. The month of May is considered to be the hottest 45^0c. While December is considered to be the coldest 23^0c in the study area contour values ranges between 300-400m. The present study area falls in the Survey of India Toposheet No: 57 J/08and57 K/05. Sample location of the study area with longitude and latitudes have been shown in Table 1 and map in figure 2.

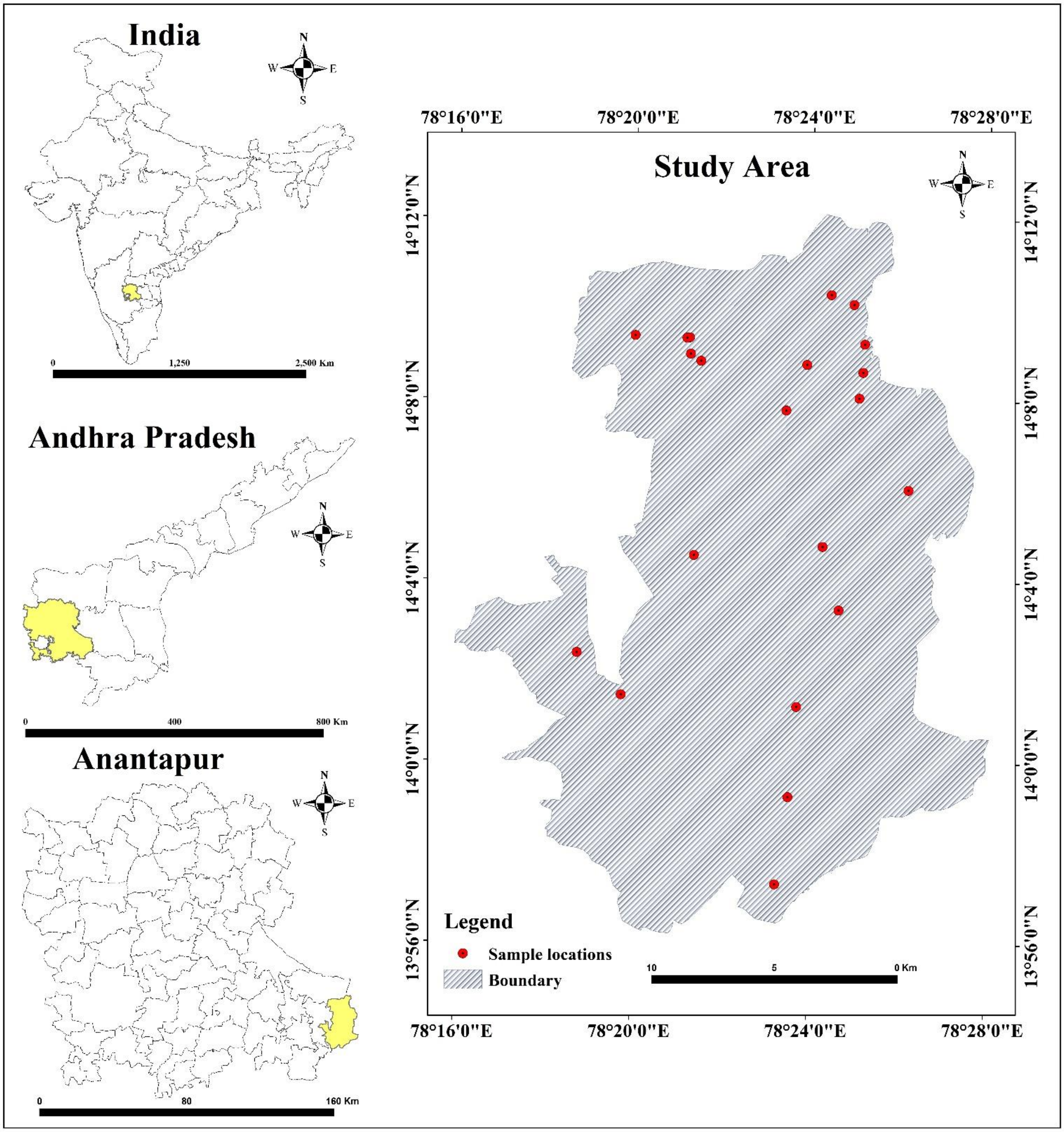

Figure 1: Location map

Table 1: Water sample locations

S.no	Name of the Village	Latitude Value	Longitude value
1	PEDABALLI	N 13.98736	E 78.39246
2	BALIJAPALLI	N 13.95516	E 78.3878
3	P.KOTHAPALLI	N 14.02058	E 78.39538
4	NAMBULAPULAKUNTA	N 14.05625	E 78.41101
5	VANKAMADDI	N 14.07956	E 78.40471
6	DANIYANICHERUVU	N 14.07615	E 78.3562
7	MEKALACHERUVU	N 14.06	E 78.3124
8	GOOTIBYLU	N 14.02458	E 78.3292
9	EDURUDONA	N 14.16556	E 78.8049
10	GOPALAPURAM	N 14.15686	E 78.33329
11	VELICHELIMALA	N 14.16819	E 78.04539
12	WEST NADIMPALLI	N 14.15611	E 78.35287
13	DEVARAPALLI	N 14.15023	E 78.35427
14	KONDAVANDLAPALLI	N 14.14763	E 78.35818
15	THATIMANUGUNTA	N 14.12965	E 78.39053
16	RANGANAPALLI	N 14.14653	E 78.39815
17	GOWKANAPALLI	N 14.17228	E 78.40713
18	DINNAMEDHAPALLI	N 14.16877	E 78.41577
19	EAST NADIMPALLI	N 14.15422	E 78.41995
20	KORUVANDLAPALLI	N 14.14383	E 78.41937
21	JOWKULA	N 14.13429	E 78.41803
22	MARIKOMMADINNE	N 14.10057	E 78.43687

Geology

The study area chiefly consisted of Kadiri schist belt. It is a linear greenstone belt situated in the eastern part of the Dharwar carton and north-east direction part of the Cuddapah basin. The schist belt comprises predominantly acidic volcanoes with a small amount of acid volcanic belonging Dharwar supergroups (Archaean age). It is covered with Migmatites and granites of the peninsular gneissic complex. Rhyolite, Rhyodacite, quartz - feldspar porphyry, Muscovite-Sericite schist, and Quartz-Sericite schist represent acid volcanic. Metabasalt and Metagabbro (Amphibolites) constitute the basic volcanic in addition; persistent bands banded iron formation (BIF) within acidic volcanic occurs the as a small amount of conglomerates (Pradeep Kumar et al., 2019). In persistent but conspicuous auto clastic conglomerate horizon occurs in conformable relationship with acid volcanic of the belt (Figure 3).

Geomorphology

Geomorphology includes the study of landforms, their processes, form, and sediments at the surface of the Earth (and sometimes on other planets). Due to the geological agents like air, water, and ice can mold the landscape. The landform is formed by its composition depth of weathering structural frame and the environment which includes soil cover, hydrology, and hydrogeology. The landforms are classified on the basis of mode of origin, relief slope factor, and surface cover (Pradeep Kumar et al., 2020). The landforms occurring in the area are grouped as Pedi plain residual hill, and structural hill shown in the figure 4.

Hydromorphology

Hydrological (water flow, energy, etc.) and geomorphological (surface features) developments and attributes of rivers, lakes, estuaries, and coastal waters. The water framework directive (WFD) dictates that the ecology of surface waters is protected by correctly managing their hydrology and geomorphology. Hydro morphology assessment and prediction requires an evolution of aquatic habitat composition and dynamic nature of water bodies and facilitates sustainable restoration.

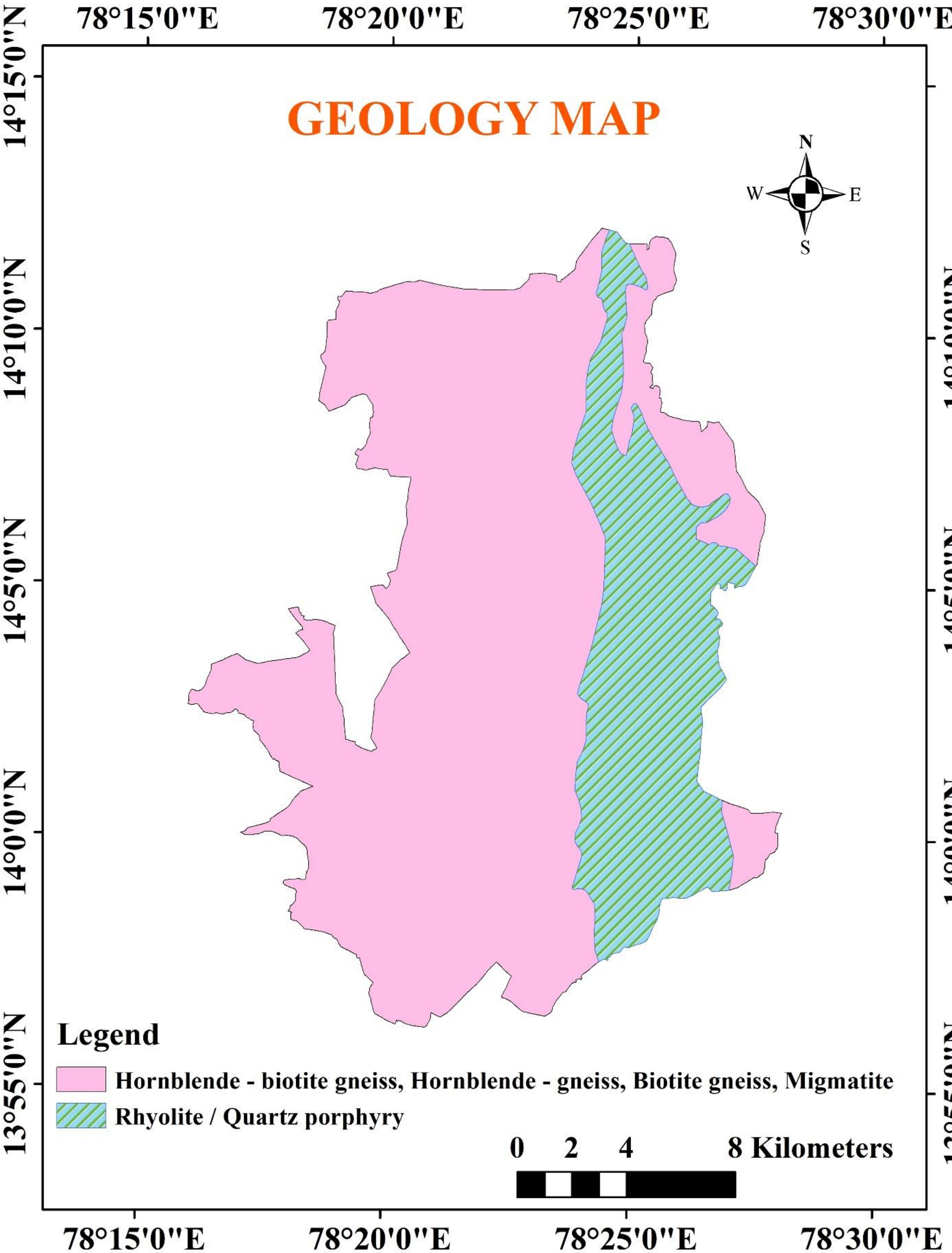

Figure 3: Geology map of the study area.

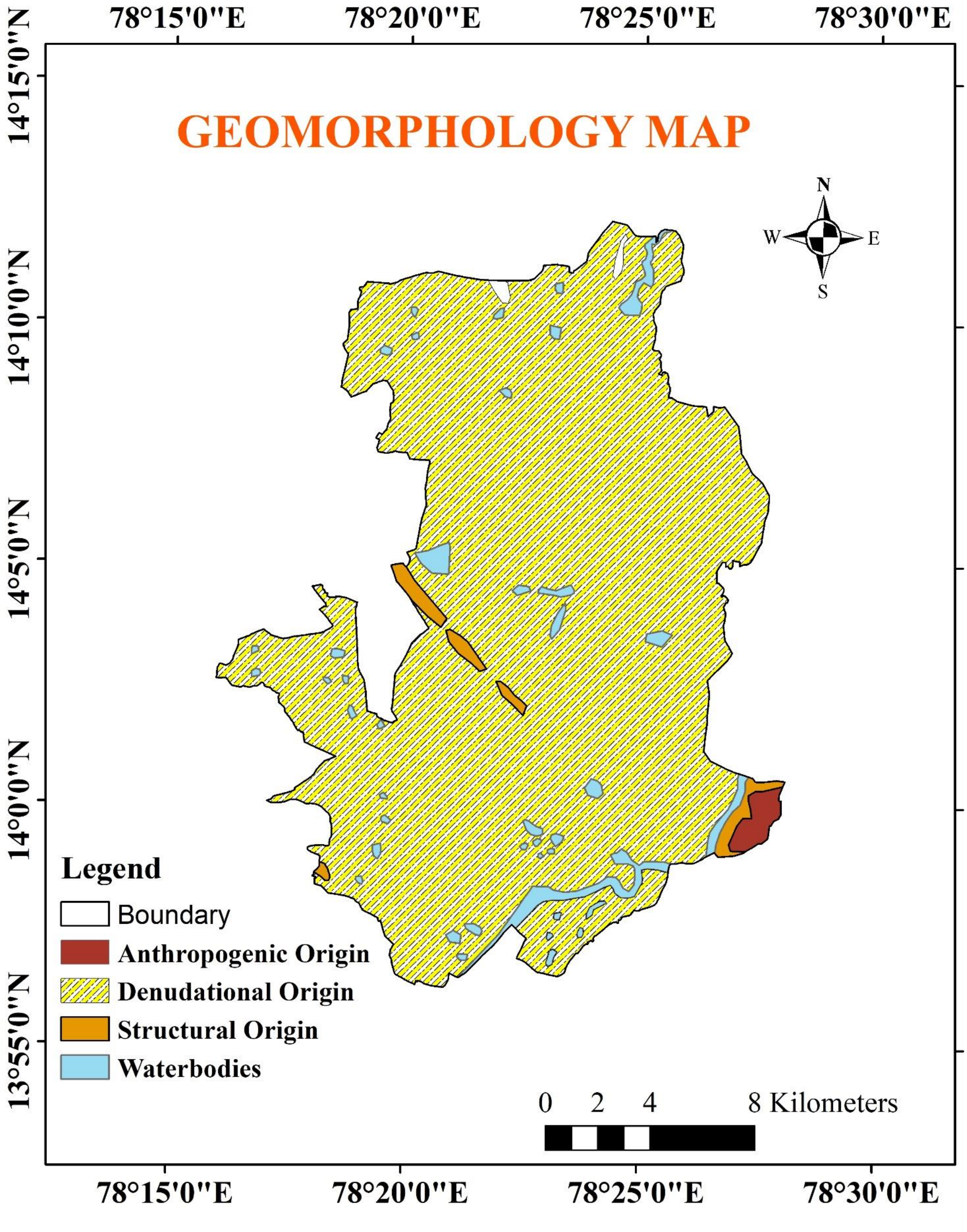

Figure 4: Geomorphology of the study area

METHODOLOGY

In order to assess the groundwater chemistry, a total of 22 groundwater samples were collected from open and bore wells of different villages of Nambulapulakunta Mandal of Anantapur District during post-monsoon season in the year 2019. This season was selected because in this season often contamination increases due to low dilution and this tends to the accumulation of ions. Before sampling, the water left to run from the source for few a minutes. The groundwater sample was collected in well cleaned 1000ml polyethylene bottles. The latitude and longitude of the sample locations are shown inTable1.These collected groundwater samples were analyzed for pH, electrical conductivity (EC), using pH and EC meters; Total dissolved solids (TDS), which were computed by multiplying the EC with a factor(0.55 to 0.75) that depend on the relative concentrations of ions; Total alkalinity (TA) as CaCO3 and bicarbonate HCO_3^- were estimated by titrating with HCl; Total hardness (TH) as $CaCO_3$ and Calcium(Ca^{2+}) were analyzed titrimetrically, using standard EDTA; Magnesium (Mg^{2+}) was calculated on the basis of the difference in the concentration between TH and Calcium Ca^{2+}Sodium(Na+)and Potassium(K^+) were measured by flame photometry; chloride (Cl-) was estimated by standard AgNO3 titration; The samples were analyzed to assess various physicochemical parameters according to APHA, 2007.

RESULTS AND DISCUSSION

Physico – Chemical parameters

Physico – Chemical parameters are very crucial and vital to the test of water because of suitable or not for purposes of drinking, household, irrigation, and industrial. Hence, collected water samples are required to be tested with various types of physic-chemical parameters.

Hydrogen Ion Concentration (pH)

pH is mainly significant for determining the acidic nature of water. Lower the pH value higher is the corrosive nature of water. pH was positively correlated with electrical conductivity (EC) and total alkalinity (Y. Ravi Kumar et al 2016). The permissible limit of pH value for drinking water 6.5 to 8.5. The pH values ranges from 7.5 - 7.9 with an average of 7.7. The pH of the study area is slightly alkaline (Nagaraju et al, 2014)

Total Hardness

The total hardness is defined as the sum of calcium and magnesium concentrations, both expressed as $CaCO_3$ in mg/L. Hardness is the main source divalent cations of calcium (Ca^{2+}), magnesium (Mg^{2+}), and, alkaline earth metal like strontium, iron, manganese, etc. Temporary Hardness is formed by mainly carbonates and bicarbonates of calcium and magnesium and permanent hardness is formed by Sulphates and chlorides (Standard et al., 2012). The TH values range from 60 to 350 mg/l with an average of 1226 mg/. The degree of hardness consumers consider objectionable depends on the degree of hardness to which consumers have become accustomed, as described here:

Soft	0 to75 mg/L as CaCO3
Moderate	75 to 150 mg/L CaCO3
Hard	150 to 300 mg/CaCO3
Very Hard	Above 300 mg/CaCO3

Electrical conductivity

The permissible limit of conductivity in groundwater should between 700 and 3000µS/cm (WHO 2012). The EC values ranges from 495 - 3760 µS/cm with an average of 1226 µS/cm. This is suitable for drinking purposes (This is provided where no organic contamination and less clay material dissolved in groundwater is)

Sodium

Sodium is the majority plentiful constituent on the earth and it is broadly spread in the soils, plants, water, and foods due to the sodium bearing mineral. Sodium - ion has occurred everywhere in water reason dissolved of sodium salts. Attentiveness of sodium heart problems and cardiovascular problems in humans (Singaraja et al., 2014). The concentration of sodium less than 20mg/L is suggested for high-risk people. The permissible limit of sodium in drinking water is 200 mg/L (WHO 2012. The Sodium values range from 64-289 mg/l with an average of 179 mg/l.

Potassium

Higher attentiveness of potassium a laxative affect problem created in humans, and it causes kidney diseases or other problems such as heart disease, hypertension, diabetes, and coronary artery diseases. The permissible limit of potassium is 1–8mg/L (WHO 2011). The potassium values range from 3.1 to 8.5mg/l with an average of 4.85 mg/l.

Calcium

The calcium concentration presented in groundwater due to the dissolved calcium carbonate, sulfate, and rarely chloride. The calcium is one of the reasons for the hardness of the water. The permissible limit of calcium in drinking water is 75 mg/l (ISI, 1983). The calcium values range from 16 mg/l to 128 mg/l with an average of 87.5 mg/l.

Table 2: Physico Chemical Parameters of ground water of the Study area

S. No.	Cl	K	pH	EC	Total Hardness	TDS	CO_3	HCO_3	Ca	Mg	Na
Units	**Mg/L**	**Mg/L**		**us/m**	**Mg/L**	**Mg/L**	**Mg/L**	**Mg/L**	**Mg/L**	**Mg/L**	**Mg/L**
1	106.5	4.1	7.8	650	100	325	48	122	72	85	107
2	177.5	5.2	7.8	1732	150	865	168	170.8	96	175	146
3	63.9	3.6	7.7	831	145	414	60	183	16	98	64
4	152.6	5.2	7.7	857	100	929	144	97.6	65	121	119
5	113.6	3.1	7.8	705	160	352	84	183	78	97	96
6	85.2	3.3	7.7	804	110	402	-	207.4	70	45	138
7	149.1	5.0	7.8	901	60	447	120	244	88	186	178
8	71	6.2	7.8	495	135	247	192	207.4	82	126	182
9	124.2	5.0	7.7	979	65	491	-	219.6	65	96	148
10	71	5.2	7.8	920	350	442	84	195.2	73	185	224
11	71	3.3	7.5	3760	240	1892	-	366	98	89	217
12	248.5	5.3	7.6	2321	60	1164	-	244	118	178	194
13	195.2	6.2	7.5	516	200	1116	240	146.4	92	135	227
14	17.7	3.3	7.6	2229	90	599	60	244	80	67	167
15	64	5.3	7.6	1200	120	547	60	488	116	125	248

16	51	5.0	7.7	1004	160	668	48	195.2	128	196	173
17	71	3.3	7.9	1096	180	531	120	158.6	112	91	259
18	106	4.5	7.7	1333	100	558	96	219.6	95	89	289
19	185	4.9	7.8	1064	80	437	-	268.4	108	98	173
20	547	5.0	7.6	1114	115	558	72	146.4	94	53	186
21	284	8.5	7.8	875	215	797	-	122	96	182	276
22	227	6.2	7.6	1586	245	1160	70	183	85	86	127

The relative weight of chemical parameters

Chemical Parameters	**WHO standards**	**Study area values**
pH	6.5 - 8.5	7.5 – 7.8.
TDS	500 - 2,000	247 - 1892
Chloride	250 - 1,000	64 - 547
Total Hardness	300 - 600	60 - 350
Calcium	75 - 200	16 - 128
Bicarbonates	244 - 732	98 - 488
Magnesium	30 - 100	45 - 196

Conclusion:

The study area with reference to the WHO 1996 standards to formative the Physico-chemical parameters of the groundwater. The post-monsoon pH values are indicating basic nature. In post-monsoon, the sweetness of the water decreases when the concentrations exceed this limit and may cause gastro-intestinal irritation. TDS concentration indicates that 6o% of the ground water falls in freshwater (<1,000 ppm) type. Most of the samples felt within the acceptable and allowable limit, only a few samples felt above the WHO limit in calcium and magnesium. Na and K elements concentration was more in post-monsoon due to the surrounding Lithology like Hornblende-biotite-gneiss rocks. The study area occupies mostly Meta basalt, Metagabbro, Rhyolite, Rhyodacite, Quartz porphyry, Quartz-feldspar porphyry, and volcanic conglomerate, hence water obtained some saline nature. All samples are within limit hence; the groundwater is suitable for drinking purposes.

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